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Research paper

A network analysis of psychosocial risk factors and depression in adolescence

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ABSTRACT

Adolescence is a period marked by social, psychological and physiological change, coinciding with a peak in depression onset. Over the last few decades, incidence of depression in adolescents has increased substantially, presenting a serious public health concern. The risk factors associated with depression have mostly been studied in isolation or without consideration for differences between age groups, limiting our ability to infer how factors are associated with each other and with depressive symptomatology. In the current study, we explored the relationships between different psychosocial risk factors and depression in adolescence, using cross-sectional data collected from adolescents aged 11–18 years ($n = 202$). We estimated two network models. The first focused on depression severity and the second on core depressive symptoms (low mood and anhedonia). In both networks, roughly half of all possible edges were estimated as non-zero (52.78% and 53.32%, respectively). Emotion regulation difficulties, perceived stress and school environment were among the most central factors and showed some of the strongest associations with both depression severity and presence of individual core symptoms. The network structure was largely independent of age, but the quality of school environment was more strongly related to depression severity in older adolescents. The study expands our understanding of the multivariate associations of risk factors and depression at different ages. The findings highlight emotion regulation, stress management skills and school environment as potential prevention and intervention targets, and point towards a need to consider differences between age groups when addressing depression in adolescence.

More than one third of adolescents experience elevated symptoms of depression and, over the last few decades, the rates of depression in adolescents have seen a sharp increase, presenting a serious public health concern (Daly, 2022; Lakasing and Mirza, 2020; Liu and Kuai, 2025; Shorey et al., 2022). In children and adolescents, depression is associated with poorer academic performance, high-risk sexual behaviour and impaired social relationships (Horowitz and Garber, 2006), and early presentations contribute to a more severe and chronic progression later in life, lower life satisfaction, cardiovascular disease, premature mortality, and the presentation of other mental health disorders (Correll et al., 2017; Kim-Cohen et al., 2003; Rohde et al., 2013; Walker et al., 2015). Refining our understanding of the factors associated with the presentation of depression in adolescence is thus particularly important because it will allow for earlier identification of individuals at risk of developing the disorder and provision of more targeted prevention and intervention strategies, while informing our understanding of its aetiology.

Broadly, depression is caused and maintained by a combination of psychological, social, and biological factors (Borrell-Carrió et al., 2004; Engel, 1977; Engert et al., 2020; Schotte et al., 2006; Sullivan et al., 2000). Some of the consistently implicated psychological factors include emotion regulation (i.e., the ability to monitor, evaluate and modify emotional reactions; Berking and Wupperman, 2012; Compas et al., 2017; Siener and Kerns, 2012; Thompson, 1994; Uhl et al., 2019) and stress (Lindholdt et al., 2022; Thorsén et al., 2022). For example, children with depression exhibit greater negative affect intensity and biased interpretations of emotional situations (e.g., catastrophising and overgeneralisation), as well as poorer awareness of the experienced emotions (Ciarrochi et al., 2008; Siener and Kerns, 2012). Similarly, high levels of perceived stress are associated with both concurrent (Thorsén et al., 2022) and future (Lindholdt et al., 2022) depression in adolescents. In terms of social factors, low social support (e.g., from parents, friends, and other social relationships), worse school environment, and lower socioeconomic status are commonly implicated (Brière et al., 2013; Liu, 2003; McMahon et al., 2017; Michelini et al., 2021; Piko et al., 2009; Teychenne et al., 2010; Väänänen et al., 2014).

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Despite a good understanding of which factors are relevant in the presentation of depression, our knowledge on how the factors themselves relate to each in the context of depression is limited. For example, emotion regulation strategies may affect social functioning as well as one's ability to cope with stressful situations, contributing to the presentation of depressive symptoms via different pathways (English et al., 2012; Nolen-Hoeksema and Watkins, 2011). Similarly, the relative importance of different factors may change with age as children progress into adolescence and begin to rely more on their peers, or encounter more stressful life events (Hankin, 2015; Conley et al., 2012; Grant et al., 2006). Thus, an investigation of the contributions and interactions of different risk factors across a period characterised by marked social, psychological, and physiological change like adolescence is warranted.

When examining the onset and presentation of depression, considering the way in which depression is conceptualised is highly relevant. The clinical presentation of depression includes symptoms such as changes in appetite, lack of energy, feelings of worthlessness, concentration and sleep problems, and suicidal ideation, alongside a lack of interest or pleasure (i.e., anhedonia) and low mood. The latter two symptoms, lack of interest or pleasure and low mood, are considered core depressive symptoms. Importantly, in children and adolescents, low mood may manifest as irritability, further highlighting the need to understand specific associations between risk factors and core symptoms unique to this age group (American Psychiatric Association, 2013; Orri et al., 2018; Stringaris et al., 2013). Depression severity is typically measured with questionnaires assessing frequency of symptom presentation, and the sum scores reflecting a total burden. Even when adolescents do not meet the criteria for a depression diagnosis, endorsing more of the symptoms is associated with a decreased quality of life and increased functional impairment (Balázs et al., 2013; Bertha and Balázs, 2013; Fergusson et al., 2005). Thus, considering the full range of depression severity, including subclinical levels, is worthwhile.

At the same time, there is substantial variation in symptom profiles among people with depression (Fried and Nesse, 2015). Examining depression as a homogenous syndrome by only using sum scores or diagnosis thresholds may limit the interpretability of research findings by ignoring individual differences in the symptoms being experienced (Fried, 2017). Clinical diagnostic criteria outlined in the DSM require the presence of at least low mood or anhedonia, symptoms playing a pivotal role in the pathology of depression, for an individual to meet criteria for a depressive episode (American Psychiatric Association, 2013). However, an individual must also endorse at least four additional symptoms, such as changes in appetite or sleep patterns, in order to meet clinical criteria, suggesting myriad combinations of symptoms resulting in the same singular diagnosis. Furthermore, different depression symptoms differentially predict risk for depression or general functional impairment, and differ in their associations with risk factors. For example, in one study, symptoms like sleep and appetite problems accounted for at most 3% of the variability in general functioning, while low mood and interest loss/anhedonia accounted for 20.9% and 13.1%, respectively (Fried and Nesse, 2014). The latter two have also been found to be the most central symptoms in a network analysis looking at depressive symptoms and risk factors in adults (Yang et al., 2026). Characterising the associations between these core symptoms and risk factors would thus allow for a complementary insight into the dynamics governing the presentation of depression, ultimately contributing to a better mechanistic understanding of the disorder.

One way of characterising both the interdependent nature of risk factors as well as the symptomatic heterogeneity in depression is by using network analysis. In a network model, the variables are represented as nodes and the unique pairwise associations between them as edges in the network (Borsboom and Cramer, 2013; Borsboom et al., 2021; Epskamp et al., 2018b). Adopting a network perspective, mental health can be understood as a complex system of interacting psychological, social and biological factors, and network analysis provides powerful insights into these complex multivariate systems (Barabási, 2012; Epskamp et al., 2018b). Past network analysis research mainly examined the associations between depression or internalising disorder symptoms, was conducted in adults, or did not consider potential moderating effects of age on network structure (Boschloo et al., 2016; Fried and Nesse, 2014; Gossage et al., 2022; Fritz et al., 2018; Kim et al., 2021; Mullarkey et al., 2019; Yang et al., 2026). Therefore, research looking at the differences at different ages in adolescence in the multivariate associations between depression and risk factors or the relationships between symptoms and risk factors is valuable.

1. The current study

Using data from a larger project (pre-registered at https://osf.io/uaf63), the current study was designed with two main goals in mind: (1) to characterise the risk factor and depression severity network structure in a community sample of adolescents, and examine how the network structure may differ across age groups within adolescence, and (2) to examine whether individual core symptoms of depression (i.e., low mood and anhedonia) show unique associations with the risk factors, and examine how the network structure may differ across age groups within adolescence. We expected emotion regulation to be highly influential relative to other risk factors in the network. We further expected that parent relations would become less connected and peer and social relationships would become more connected to the remaining risk factors with increasing age, due to adolescents' increasing independence from parents and reliance on peers. Last, we expected that the relationships of the two depression symptoms with the risk factors would differ, where low mood, compared to anhedonia, would be more strongly associated with emotion regulation and less strongly with physical wellbeing.

2. Methods

2.1. Participants and recruitment

Participants were recruited primarily via secondary schools in England between May 2024 and August 2025, as well as through community-based organisations and local libraries. In some cases, schools contacted the research team by responding to an advert on the lab website. In other cases, schools were contacted by researchers directly via email or phone call. Information sheets about the study were shared with the school staff, who distributed them to students and their parents. Interested participants (and their parents if participants were under 16) then had two weeks to complete an online consent form via a secure and encrypted website and indicate their interest in taking part. Once participants provided consent, an in-person information and onboarding session lasting approximately 1 h was organised at the school, where a researcher introduced the study and the materials (i.e., smartphone app and login process) to the participants. For the participants who were not recruited via a school, an online information session covering the same content was organised. The study was reviewed and approved by the University College London Research Ethics Committee (Ethics Project ID number: 22135/003).

Data were collected via the eMoodie app (https://emoodie.com) where participants completed a battery of 12 mental health questionnaires independently, at home in their own time and on their own personal devices within 2 weeks of the information and onboarding session. The eMoodie interface was ‘gamified’ such that participants were incentivised with points to complete as many questionnaires as possible. Variables included in the models were selected based on prior research and were preregistered at https://osf.io/zwspj/files/3shbx. Consideration was given to the number of variables included in the model based on the sample size, with the intention of maximising sensitivity and precision of model estimates (Huth et al., 2025; Isvoranu and Epskamp, 2023).

A total of 253 adolescents took part in the study. 202 participants

completed all of the relevant questionnaires and were included in the final analysis. No additional exclusion criteria based on data or participant characteristics were applied. Participants with missing data did not differ from the final sample in terms of depression severity, age or sex (see Supplementary Materials Table 1). The sample was mostly White (57.43%), followed by Asian (19.80%), Mixed (9.90%), Black (8.91%) and other (2.48%), with 1.49% not having provided a response. Most participants were assigned female sex at birth (85.64%, $n = 173$). The age range was 11–18 years, and the average age was 15.45 ± 1.74 years. 21.78% ($n = 44$) of participants reported receiving some kind of mental health support at the time of taking part in the study.

2.2. Measures

The 8-item Patient Health Questionnaire (PHQ-8) was used as an indicator for overall depression severity, and also to assess the endorsement of individual core symptoms of depression – low mood and anhedonia (Kroenke and Spitzer, 2002). The questionnaire includes items asking about the frequency of being bothered by depressive symptoms in the preceding two weeks (the suicidality item, present in the PHQ-9, was omitted for safeguarding reasons). Participants responded on a scale from 0 (not at all) to 3 (nearly every day). The overall severity measure was obtained by summing the item-level responses, with higher scores indicating greater severity. For the second (symptom) model, symptom endorsement was translated into a binary variable, where a score of 0 on the low mood and anhedonia items indicated 'absent', whereas any score of 1 or above indicated 'present'. The questionnaire has been shown to have good psychometric properties for identifying depression and assessing depressive symptoms, and good validity and internal consistency (Cronbach's alpha = 0.89, (0.81 in the current sample); Kroenke et al., 2001, 2009; Shin et al., 2019; Wu et al., 2020).

The Difficulties in Emotion Regulation Scale (DERS) is a 36-item scale that asks respondents about how they relate to their emotions (Gratz and Roemer, 2004). The questionnaire measures difficulties in emotion regulation along six dimensions: (1) non-acceptance of emotional responses (secondary responses to negative emotions and non-accepting reactions to distress), (2) difficulties engaging in goal-directed behaviour subscale (difficulties in concentrating and accomplishing tasks when experiencing negative emotions), (3) impulse control difficulties (difficulties remaining in control of behaviour when experiencing negative emotions), (4) lack of emotional awareness (inattention to and lack of awareness of emotional responses), (5) limited access to emotional regulation strategies (belief that little can be done to regulate emotions effectively when upset), and (6) lack of emotional clarity (degree to which individuals know and understand the emotions they experience). Items include statements like 'I pay attention to how I feel' and are rated on a 5-point Likert scale from 'Almost never' to 'Almost always'. For the network analysis, the total sum score was used. The questionnaire shows strong psychometric properties across diverse populations and has been used in adolescent populations. It has good validity and internal consistency (Cronbach's alpha = 0.93 (0.95 in the current study), Cronbach's alpha for the subscales = 0.80–0.89 (0.81–0.91 in the current sample); Fowler et al., 2014; Gratz and Roemer, 2004; Neumann et al., 2010).

The 10-item Perceived Stress Scale (PSS) explores how different situations affect respondents' feelings and perceived stress (Cohen, 1988). Participants rate statements about their experience of stress in the last month such as '...been upset because of something that happened unexpectedly' on a scale from 0 (never) to 4 (very often). In the analysis, the sum score was used. The scale shows good validity and internal consistency (Cronbach's alpha = 0.88 (0.82 in the current sample); Lee, 2012; Whitney et al., 2022).

An estimate of participants' socioeconomic status was obtained with the Family Affluence Scale version 3 (FAS; Currie et al., 1997; Torsheim et al., 2016). This 6-item asset-based scale contains items like 'How many computers does your family own?'. The item responses are summed, producing a total score ranging between 0 and 13. The scale has been used extensively to examine family material conditions of children and adolescents, is an important predictor of young people's health, has good criterion validity and moderate internal consistency (Cronbach's alpha = 0.61 (0.64 in the current sample); Currie et al., 2024; Elgar et al., 2016). For the network analysis, the sum score was used.

The 27-item KIDSCREEN-27 questionnaire was used to measure generic health-related quality of life of children and adolescents, emphasising perception of psychosocial aspects of being (KIDSCREEN Group Europe, 2006). Four dimensions of the KIDSCREEN-27 were included in the analysis: 1) physical wellbeing (e.g., feeling fit, energetic, being physically active), 2) autonomy and parent relations (e.g., having enough time for yourself, your parents having enough time for you, having enough money), 3) peers and social support (e.g., spending time with friends and being able to rely on them), 4) school environment (e.g., being happy at school and able to pay attention, getting along with the teachers). The KIDSCREEN-27 also measures psychological wellbeing (e.g., being in a good mood, having fun, not feeling lonely), but this subscale was not included in the analysis due to its high content overlap with PHQ-8. Participants respond on a 5-point Likert scale ranging from 'not at all' to 'extremely', or 'never' to 'always' based on their experiences in the past week. Previous studies have shown the questionnaire to be psychometrically robust and to have good internal consistency (Cronbach's alpha 0.78–0.84 for individual dimensions (0.77–0.89 in the current sample); Robitail et al., 2007).

2.3. Statistical analyses

2.3.1. Estimating the networks

Analyses were conducted in R version 4.4.0 (R Core Team, 2023). To fit the network models, we used the mixed graphical model which is implemented in the R package *mgm* (version 1.2.14, Haslbeck and Waldorp, 2020) and allows for the inclusion of binary variables as well as moderators. The R package *qgraph* (version 1.9.8, Epskamp et al., 2012) was used to visualise the networks.

The mixed graphical model is formed by fitting multiple penalised generalised linear models, one for each node, where a given node is predicted by all the remaining nodes in the model. Thus, any given pair of node will have two associated edge estimates, one for X_i as the outcome and X_j as one of its predictors, and one for X_j as the outcome and X_i as one of its predictors. The model uses ℓ_1-regularised (LASSO) regression to enforce sparsity (i.e., the assumption that most of the parameters in the true model are equal to zero) by shrinking edge weights and setting small weights to zero. Cross-validation with 10 folds was used to find the optimal penalisation parameter λ, which controls the strength of the penalisation (Haslbeck and Waldorp, 2020). The edge estimates relating to a given pair of nodes X_i and X_j were combined according to the 'AND' rule, meaning that they were only combined (i.e., averaged) and shown in the network if both were estimated as non-zero. No additional thresholding was applied to allow for better comparisons across networks. Conceptually, any edge between a given pair of nodes suggests that they are associated with each other after controlling for all remaining nodes, and the lack of an edge between two nodes represents conditional independence (Burger et al., 2023). Before fitting the models, we checked whether any of the nodes may be redundant with each other. Using the *goldbricker* function from the *networktools* package (version 1.6.0; Jones, 2025), we observed that no pairs of variables correlating at 0.7 or above (mean $|r| = 0.34$; only perceived stress and emotion regulation crossed this threshold with 0.73) differed in terms of their correlations with the remaining variables in less than 25% of cases.

We estimated two models. In the first model, the main variable of interest was depression severity, measured as the sum (total) score on the PHQ-8. The risk factors included were difficulties in emotion regulation (DERS total sum score), perceived stress (PSS total sum score), socioeconomic status (FAS total sum score), and physical wellbeing,

autonomy and parent relations, peers and social support, and school environment (all indexed as corresponding subscale sum scores of the KIDSCREEN-27). Age was included as a moderator. In this model, all variables were continuous and coded as such.

In the second model, the same model fitting procedure was followed and the same risk factors as well as age as a moderator were included, but instead of a sum-score representing depression severity, the two core symptoms of depression of low mood ('Feeling down, depressed, irritable or hopeless') and anhedonia ('Little interest or pleasure in doing things'), coded as binary present/absent (with a score of greater than 0 indicating 'present'), were included to address symptom heterogeneity in depression. The symptom scores were dichotomised as using the raw ordinal item scores (4-point Likert) and treating them as continuous may lead to decreased reliability of the edge estimates (Boschloo et al., 2016; Fried et al., 2015; Liddell and Kruschke, 2018; Mullarkey et al., 2019). An additional exploratory sensitivity analysis with the original symptom scores was performed and the full results are presented in the Supplementary Materials (Tables 6 and 7).

The networks were examined in terms of the edges relating to depression severity and depression symptoms, as well as in terms of their topology, that is, measures capturing the structure of the network. The centrality indices are reported in terms of their standardised values (z-scores). To examine differences in network structure at different ages, we included age as a moderator in both models. Thus, a three-way interaction effect between age and each of the edges was obtained (see Haslbeck et al., 2021 for more information). The interaction effects were visualised by conditioning the network on a specific value of the moderator (i.e., one standard deviation above and below the mean age) and displaying the obtained edge estimates.

2.3.2. Stability of the networks and the centrality metrics

The stability and accuracy of the edge estimates and moderation effects of age were evaluated via 1000 nonparametric bootstraps as implemented in the *mgm* package (Haslbeck & Waldorp, 2020) and permutation testing (Epskamp et al., 2018a). In the permutation testing, we focused on the edges relating to depression severity, the two depressive symptoms, and moderating effects of age. We created 1000 permuted datasets for each of these variables and calculated the proportion of instances in which an edge of equal or greater magnitude as the corresponding edge in the original network was observed in the permuted datasets, thus obtaining empirical p estimates evaluated at $\alpha = 0.05$ for each of the specified variables (Delamain et al., 2026).

We tested the stability of the centrality measures (i.e., topology metrics) for each of the nodes via a casedropping bootstrap procedure, using the correlation stability (CS) coefficient as implemented in the *bootnet* package (version 1.6, Epskamp et al., 2018a). The CS coefficient represents the maximum proportion of cases that can be dropped from the sample while retaining a correlation of at least 0.7 between the ordering of original centrality measures and the centrality measures from the subsets with 95% probability. The CS coefficient should not be below 0.25 and ideally above 0.5 (Epskamp et al., 2018a).

3. Results

The means, standard deviations and normality metrics of the variables included in the network models are presented in Table 1. Overall, the variables were only slightly skewed and approximately normally distributed. The assumption of multivariate normality was checked with *networktools* package (version 1.5.2). Given significant deviations from multivariate normality, variables with a moderate or greater skew ($>|0.50|$; see Table 1) were deemed to violate the normality assumption (PHQ-8, age and peer and social relationships) and transformed using a nonparanormal transformation as implemented in the *huge* package (version 1.3.5; Liu et al., 2009).

Despite being a convenience-based sample, the average PHQ-8 score in the sample was relatively high at 9.99 (scores $\geq$10 indicate moderate clinical depression; Kroenke et al., 2009) and did not significantly differ between male and female participants ($t(43.86) = 0.47$, $p = 0.637$). In terms of the two core depression symptoms, 172 (85.15%) participants reported feeling down, depressed or hopeless for at least several days in the two weeks preceding data collection, while 156 (77.23%) reported little interest or pleasure in doing things.

Table 1
Descriptive statistics of variables included in the network models.

Variable	Mean	SD	Min	Max	Skew	Kurtosis
Depression severity (PHQ-8)	9.99	5.20	1	24	0.51	−0.46
Age	15.45	1.74	11	18	−0.79	−0.18
Socioeconomic status	8.74	2.52	3	13	−0.25	−0.70
Perceived stress scale	23.61	5.96	9	38	−0.03	−0.29
Difficulties in emotion regulation	104.54	26.93	55	166	0.20	−0.90
Physical wellbeing	13.88	3.55	5	23	0.17	−0.26
Autonomy and parent relation	24.05	5.47	7	35	−0.20	−0.41
Peers and social support	14.79	3.81	4	20	−0.65	0.00
School environment	13.03	3.52	4	20	−0.21	−0.38

3.1. Model 1: depression severity

3.1.1. Network structure

The network of depression severity and risk factors is shown in Fig. 1a. The mean absolute weight of edges was 0.09, and 19 of the possible 36 (52.78%) were estimated as non-zero. Depression was positively associated with difficulties in emotion regulation and perceived stress, and negatively with school environment quality and physical wellbeing. Depression severity had a predictability score (R^2) of 0.58, indicating that more than half the variance was explained by the nodes it shared edges with (for other predictability scores, see Supplementary Materials Table 2).

Age emerged as a significant moderator of several edges in the network (Fig. 1b). With increasing age, depression severity and school environment showed a stronger negative association. At the same time, age had a negative moderating effect on the positive association between physical wellbeing and autonomy and parent relation, and on the positive association between peers and social support and school environment. Physical wellbeing and peers and social support showed a change from a positive relationship below mean age to a negative relationship above mean age; however, the association itself remained small. Age is not connected to any of the nodes in Fig. 1b because the three networks are conditioned upon specific values of age (mean, one standard deviation below and one standard deviation above).

3.1.2. Network topology

In terms of strength (i.e., the sum of the absolute value of the edges of a node), perceived stress ranked highest, followed by school environment, difficulties in emotion regulation and depression severity (Fig. 2a), meaning that these nodes had a relatively strong associations with other nodes in the network. In terms of age-related difference in the centrality estimates of the nodes, the autonomy and parent relation node was less central, relative to other nodes, in older participants, which seemed to be driven by the reduced association with physical wellbeing. Depression severity was also more central in older participants, reflecting the stronger negative association with school environment (Fig. 2b).

3.1.3. Stability testing

The pairwise effects (corresponding to edges) showed relatively small variance across bootstraps, indicating that the estimates are stable. Mean estimates of the moderation effects were relatively small but still stable, as indicated by a low variance. The resulting average estimates and the 95% quantiles of each of the edges and moderation effects of age

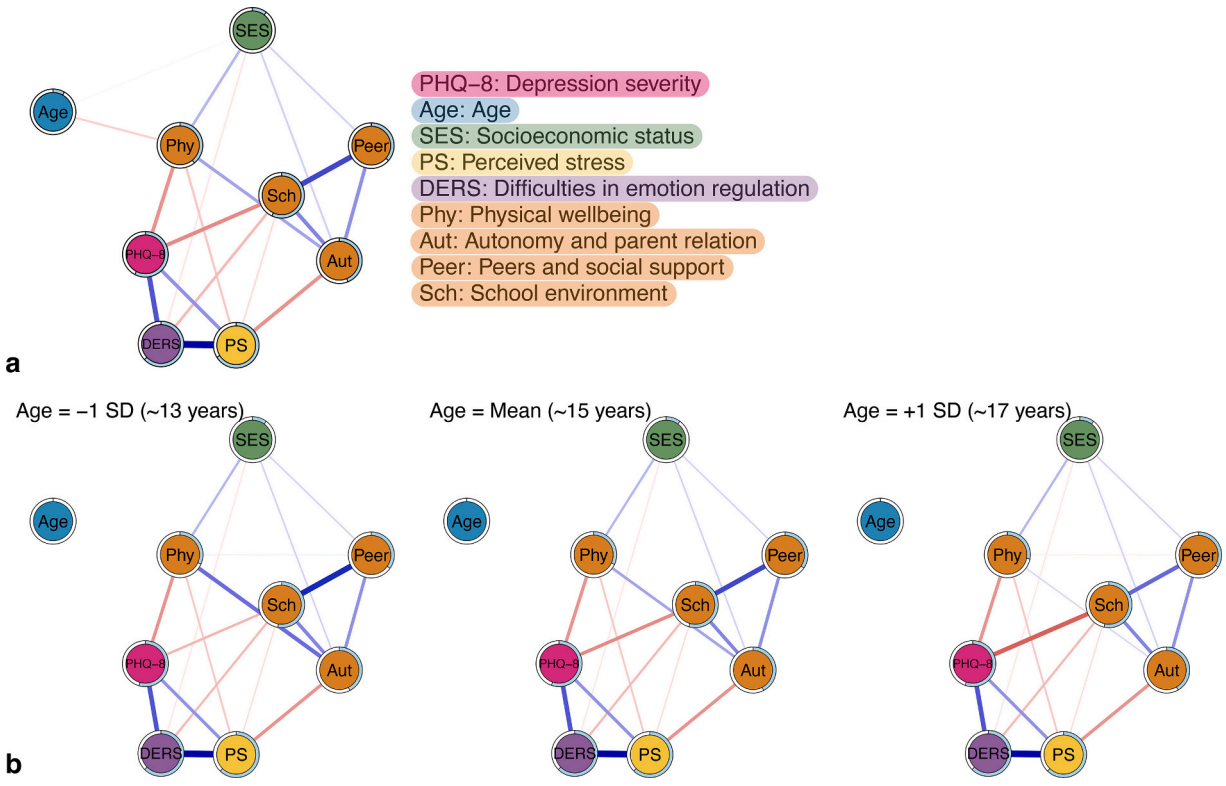


Fig. 1. Depression severity network models.
(a) The depression severity network model. Blue edges indicate positive associations, red edges indicate negative associations. Thickness and saturation of the edges correspond to the strength of the association. Pie charts around the nodes show the predictability of each node by the remaining nodes in the network. Edge-weight estimates ranged from 0.01 to 0.47. (b) The depression severity network model with age as a moderator, conditioned on age at 1 standard deviation below mean, mean, and 1 standard deviation above mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

are shown in Supplementary Materials Fig. 1.

The proportion of permuted samples in which estimates of a relationship as strong as or stronger than that observed in the original dataset was smaller than 0.05 for all edges relating to depression severity, and all moderation effects of age (Benjamini-Hochberg FDR-corrected *p*-values all remained significant; see Supplementary Materials Table 3).

Last, we evaluated the topology measures of the network. While betweenness (CS(cor) = 0) and closeness (CS(cor) = 0.00) were unstable, strength showed excellent stability (CS(cor) = 0.59), indicating that approximately 59% of observations could be dropped and the order of nodes in terms of strength would still correlate at least 0.7 with that of the original network (see Supplementary Materials Fig. 2).

3.2. Model 2: symptoms

3.2.1. Network structure

The network including the two core depression symptoms (low mood and anhedonia) and risk factors is shown in Fig. 3a. The mean absolute weight of edges was 0.09, and 24 of the possible 45 (53.32%) were estimated as non-zero. The presence of both low mood and anhedonia was associated with greater difficulties in emotion regulation and lower physical wellbeing. However, only low mood was associated with increased perceived stress and decreased peers and social support scores, while only anhedonia was significantly associated with lower school environment scores. The classification accuracies for low mood and anhedonia were 0.88 and 0.82, respectively, and thus slightly higher than their marginal probabilities (0.85 and 0.77). The variances explained for each of the remaining nodes are reported in Supplementary Materials Table 4.

Moderation effects in this network were similar to those observed in the depression severity network. Namely, age had a negative moderating effect on the positive relationships between physical wellbeing and autonomy and parent relation, and peer and social support and school environment. Alongside this, the relationship between physical wellbeing and peers and social support was positive in younger age groups but negative in older groups. Moreover, the positive association between physical wellbeing and socioeconomic status was slightly weaker at older ages. Socioeconomic status and anhedonia reversed from a small positive to a small negative association at older ages (see Fig. 3b). Being the moderator, age is not connected to the remaining nodes in the network in Fig. 3b.

3.2.2. Network topology

Overall, based on strength, perceived stress had the greatest influence on the network structure, followed by difficulties in emotion regulation, low mood and school environment (Fig. 4). In terms of age-related differences in the strength estimate of the nodes, perceived stress, difficulties in emotion regulation and low mood consistently showed the strongest relationships with other nodes in the network.

3.2.3. Stability testing

The estimates of pairwise effects were less stable than in the first model as indicated by a greater variance in the estimates, especially for estimates relating to the binary symptom nodes. As in the first model, the mean estimates of moderation effects were relatively small but generally stable. The stability of the parameter estimates is reported in Supplementary Materials Fig. 3.

Regarding the permutation testing, for both symptoms and all the moderating effects of age, the estimated edges were unlikely to have

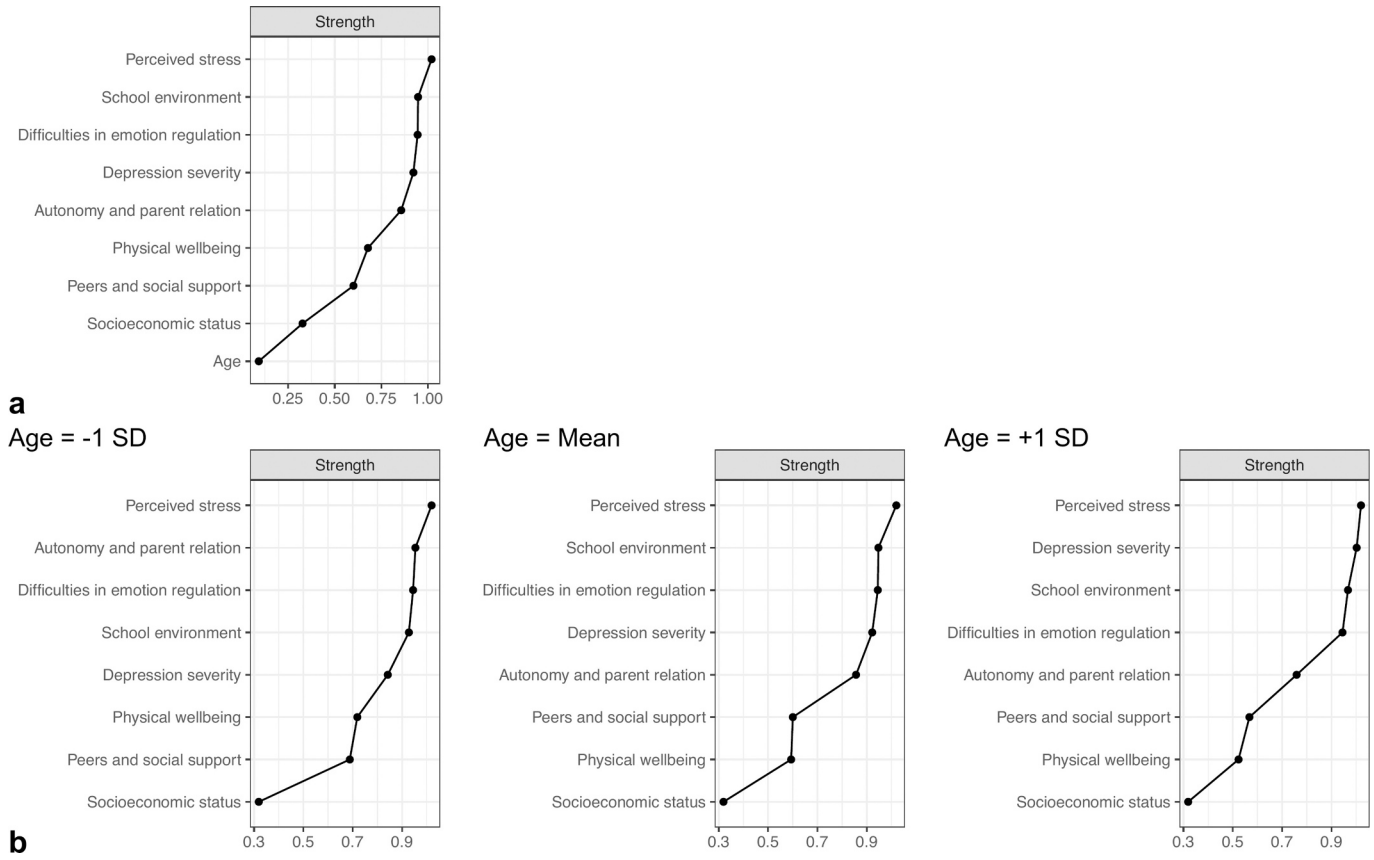


Fig. 2. Strength scores for the nodes in the depression severity network model.
(a) Strength estimates (x-axis) of each node, i.e., the sum of the absolute edge weights for each node. (b) Strength estimates of each node for networks estimated at one standard deviation below the mean, mean and one standard deviation above the mean of age.

been observed due to chance (all empirical p values and Benjamini-Hochberg FDR-corrected p-values <0.05; Supplementary Materials Table 5).

Looking at the centrality measures, the *CS*-coefficient for betweenness (*CS*(cor) = 0.00), closeness (*CS*(cor) = 0.00), and strength (*CS*(cor) = 0.13) were below the lower threshold of 0.25, indicating the measures were not stable. Thus, the measures were not further interpreted (see Supplementary Materials Fig. 4).

3.3. Exploratory analysis

We ran two linear regression models looking at the specific role of difficulties in emotion regulation and school environment as they relate to depression severity, given these variables' consistent implication across networks. Specifically, we looked at the subscales of the DERS scale and the items of the school environment subscale from the KIDSCREEN-27 questionnaire in two separate models to examine which aspects of emotion regulation and the school environment are most strongly related to depression severity. In both models, we controlled for all the risk factors included in the network models, age, and sex, and included age interaction terms for the DERS subscales (first model) and KIDSCREEN-27 school environment subscale items (second model).

The first model (DERS subscales) indicated that the predictors explained 54.23% of the variance in depression severity ($R^2_{adj} = 0.54$, F(20, 181) = 12.91, $p < 0.001$). Of the DERS subscales, lack of emotion clarity ($\beta = 0.16$, $p = 0.017$) was significantly associated with depression severity. None of the remaining subscales or their interactions with age were significant predictors of depression severity ($p > 0.05$).

The second model (school environment subscale items of the KIDSCREEN-27) explained 56.50% of the variance in depression severity ($R^2_{adj} = 0.56$, F(16, 185) = 17.32, $p < 0.001$). In terms of the items comprising the school environment subscale of the KIDSCREEN-27 questionnaire, being able to more frequently pay attention at school was significantly associated with lower PHQ-8 scores ($\beta = -0.16$, $p = 0.026$), while the remaining items or their interactions with age were not significant predictors of depression severity ($p > 0.05$).

3.3.1. Sensitivity checks

Given the large proportion of participants excluded due to missing data, we performed multiple imputation, obtaining 12 imputed datasets (van Buuren & Groothois-Oudshoorn, 2011; White et al., 2011). We fit each of the network models (model 1 and model 2) on each of the datasets and compared the results presented above to the ones obtained in the imputed datasets. For model 1 (depression severity), the average Spearman rank correlation of the edge weights between the original network and networks obtained in each of the imputed datasets was very strong (mean r_S(SD) = 0.95(0.02), min = 0.91, max = 0.97). The average correlation of the moderation effects in this model was weak (mean r_S(SD) = 0.37(0.22), min = −0.04, max = 0.64). For model 2 (depression symptoms), the average Spearman rank correlation of the edge weights between the original network and imputed networks was very strong (mean r_S(SD) = 0.96(0.03), min = 0.89, max = 0.98), while the correlation of the age moderation effects was moderate (mean r_S(SD) = 0.43(0.28), min = 0.00, max = 0.89). Full details on the multiple imputation procedure and the recoverability of individual edge weights and moderation effects are reported in the Supplementary Materials – Multiple Imputation (Supplementary Materials Tables 8–11).

To examine any potential effects of dichotomising the symptoms variables in model 2 (symptom network), we also fit a network model on

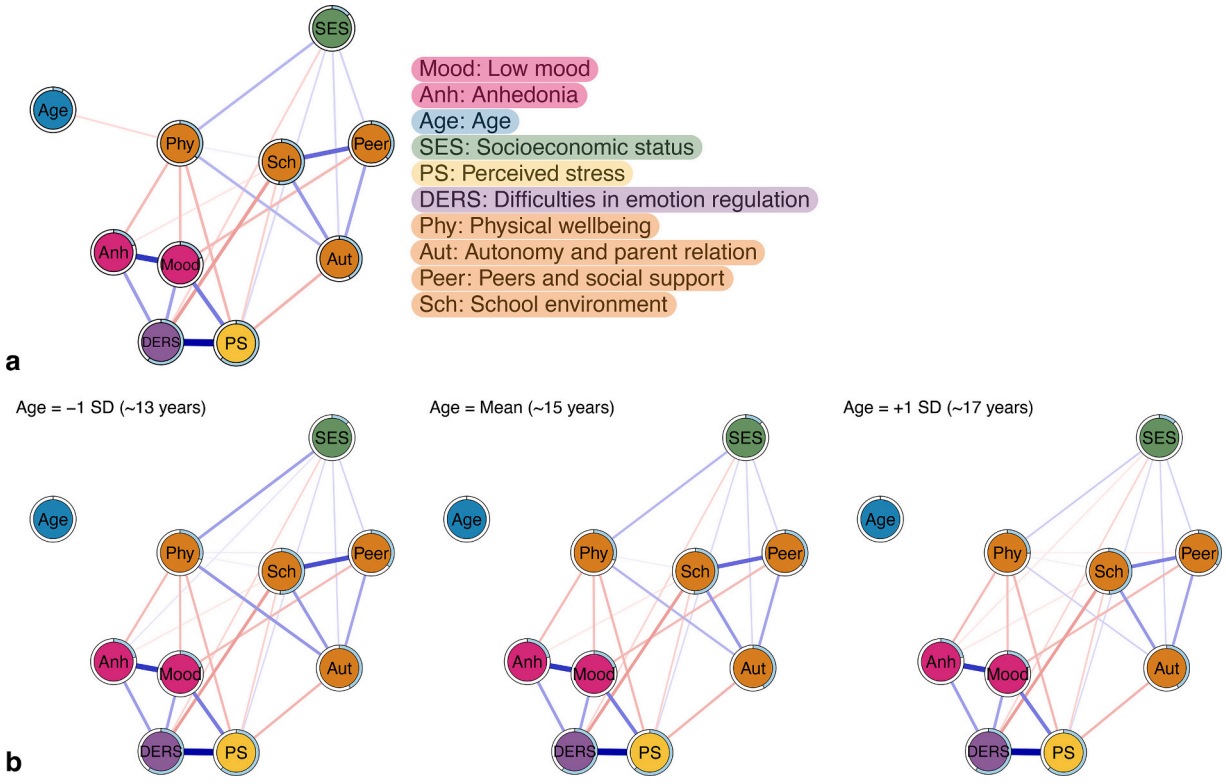


Fig. 3. Depression symptoms network models.
(a) The depression symptom network model. Pie charts around the nodes show the proportion of variance explained of each node by the remaining nodes in the network. Low mood and anhedonia are coded as binary (symptom is either present or absent) and positive (blue) associations correspond to a greater likelihood of the symptom to be present versus absent. For reference, the maximum estimated edge weight had a value of 0.54, minimum a value of 0.03. The pie chart for low mood and anhedonia show the normalised classification accuracy (i.e., how much the error in predicting the presence or absence of symptoms is reduced by the model relative to chance (Haslbeck and Waldorp, 2018)). Edge-weight estimates ranged from 0.03 - 0.54. (b) The depression severity network model with age as a moderator, conditioned upon age at one standard deviation below mean, mean, and 1 standard deviation above mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the original (complete cases) dataset using the full range of symptom scores (4-point Likert). The obtained network included all edges reported above (Fig. 3), except the edge between socioeconomic status and perceived stress, and three of the five age moderation effects, with Spearman rank correlation very strong for edges ($r_s = 0.96$) and moderate for moderation effects ($r_s = 0.58$; see Supplementary Materials Figs. 6 and 7).

4. Discussion

In this study, we used network analysis to examine how psychosocial risk factors interact and relate to depression severity and individual core symptoms in a community sample of adolescents. Results showed that depression severity, low mood and anhedonia were all related to reduced physical wellbeing and difficulties in emotion regulation. We also observed age-related differences in network structure, with school environment quality being more strongly associated with depression severity in older participants.

Regarding our first aim, which was to characterise the risk factor and depression severity network, we found that depression severity was uniquely associated with emotion regulation difficulties, stress, reduced physical wellbeing and a lower school environment quality. Results align with previous research showing that emotion regulation plays a central role in the development and maintenance of depression (Berking and Wupperman, 2012; Michelini et al., 2021) and people diagnosed with depression have been found to experience difficulties in identifying, accepting, interpreting and modifying emotions (Berking and Wupperman, 2012; Ciarrochi et al., 2008; Siener and Kerns, 2012). Indeed, we found emotion regulation to be one of the most influential risk factors in adolescence.

Depression was also strongly associated with perceived stress, which emerged as one of the most central nodes in the network. Previous work has suggested that increased levels of stress experienced by adolescents as they grow older (due to e.g., increased responsibilities at home and at school) may explain the age-dependent increase in psychopathology across this period (Compas et al., 2017; Copeland et al., 2011; Lindholdt et al., 2022; Thorsén et al., 2016, 2022; Wiklund et al., 2012). While we did not find evidence for a stronger relationship between stress and school environment at older ages, we did observe the inverse relationship between school environment and depression severity to be stronger in older participants. This may be reflecting increasing academic and social demands placed on adolescents as they approach the end of secondary school. However, it should be noted that our sensitivity analysis using multiple imputation suggested that the observed moderation effects were less reliable, possibly due to the very small and less stable effect estimates.

At the same time, our exploratory analyses indicated that the association between depression severity and school environment may be explained by participants' ability to pay attention at school. Ability to pay attention overlaps conceptually with depressive symptoms (i.e., having difficulty with concentration) and thus an alternative interpretation of the observed effect could be that depression-related concentration problems affect the experience of a school environment more as adolescents grow older and face greater academic demands. Nevertheless, a unique negative association was also observed between anhedonia and school environment as well as stress and school environment,

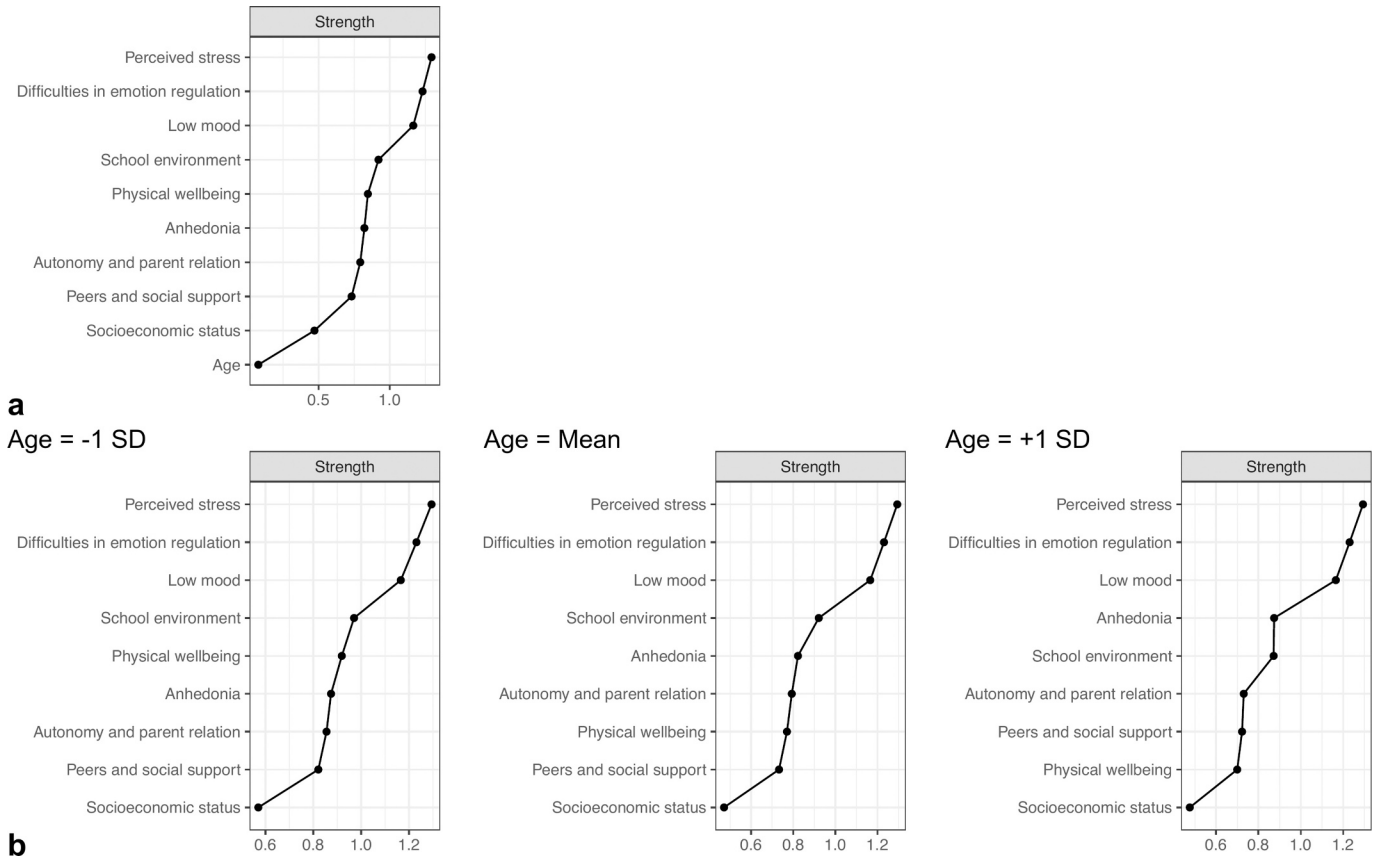


Fig. 4. Strength scores for the nodes in the symptoms network model.
(a) Strength estimates (x-axis) of each node, i.e., the sum of the absolute edge weights for each node. (b) Strength estimates of each node for networks estimated at one standard deviation below the mean, mean and one standard deviation above the mean of age.

suggesting that the school environment may play an important role in wellbeing. Providing further support to adolescents, possibly in the form of a social support network, learning opportunities, and fairness and rules, may thus be beneficial (Brière et al., 2013; Compas et al., 2017; Gossage et al., 2022; Piko et al., 2009).

Importantly, both stress and emotion regulation appear to be interrelated factors associated with psychopathology. As outlined by Nolen-Hoeksema and Watkins (2011), exposure to stress may represent a distal risk factor and its association with psychopathology may be mediated and moderated by a more proximal factor of emotion regulation. This is supported by our models which showed a positive association between all three variables (depression severity, emotion regulation difficulties and perceived stress). From this perspective, our results suggest that reducing the stress source itself, improving the coping abilities or the regulation of specific emotions, and increasing resilience would be successful methods of improving wellbeing in adolescence.

In terms of specific symptoms, we found that anhedonia, but not low mood, was associated with school environment, while low mood, but not anhedonia, was associated with perceived stress and peers and social support. Mirroring the first model, the two symptoms were also associated with emotion regulation and physical wellbeing. In line with our results, previous research in adults and adolescents has found that different adverse life events are associated with different patterns of depressive symptoms and that different symptoms are associated with different functional impairments (Fried and Nesse, 2014; Fried et al., 2014; Gossage et al., 2022; Keller et al., 2007; Lux and Kendler, 2010). For example, Fried and Nesse (2014) examined the associations of depressive symptoms with functioning in adults and found that sad mood was a strong predictor in multiple domains of functioning, including work, home management, social activities, private activities and relationships. Anhedonia showed some overlapping relationships but seemed to be most strongly related to social functioning. While the edges relating to low mood and anhedonia in our model showed low stability, likely due to the small proportion of participants not reporting any symptoms, and should be interpreted with caution, the observed unique relationships of each of the two symptoms provide support for investigating psychopathology at the symptom level in adolescence in larger samples.

4.1. *Limitations and future directions*

A limitation of the current project is its cross-sectional nature which does not allow for an investigation of causal dynamics. Future longitudinal designs would allow for establishing the temporal progression of depression across adolescence, as well as identify early predictors of future depressive symptoms which could be used to detect and treat depression early. Additionally, the sample was heavily skewed towards females, limiting the generalisability of the findings as rates of depression are notably higher in this population and, due to differences in symptom presentation, depression symptoms may also be overestimated in females compared to males (Bulhões et al., 2019; Liu and Kuai, 2025). Future studies should therefore aim to recruit more representative samples.

Second, the variables included in the network mainly correspond to questionnaire sum scores, limiting the specificity of the findings. It would be beneficial, for example, to estimate the associations between all individual depressive symptoms and individual items or subscales of the remaining questionnaires (Epskamp et al., 2018a). We aimed to address this through the exploratory subscale- and item-level analyses, but future research in a larger sample should seek to jointly model the

more fine-grained relationships. This will provide a more nuanced insight into the multivariate associations of depressive symptoms and different risk factors, which will be crucial for clarifying more precise intervention targets in young people (Borsboom et al., 2021; Epskamp et al., 2018b).

Related to this, we opted to include only the two core symptoms of depression in this study as they have been shown to be central symptoms in previous network analysis research, both represent core diagnostic symptoms, and the inclusion of all symptoms was expected to result in unstable networks in our study (Yang et al., 2026). However, appetite and weight change, energy loss, and insomnia are common symptoms of depression in adolescents and may help better explain some of the observed associations, and in different ways for different people (Rice et al., 2019). For example, anhedonia may contribute to a worse experience of a school environment due to its associations with energy loss or sleep problems. Additionally, irritability has been suggested to better reflect low mood or to be an antecedent of depression/low mood in childhood and adolescence (Rice et al., 2017; Vidal-Ribas et al., 2016), but others (Rice et al., 2019) have found no significant differences between adults and adolescents. Thus, jointly examining multiple symptoms would shed more light into the heterogeneity of depression (Fried and Nesse, 2014).

Last, while the strength centrality measure was stable and interpretable in the depression severity network, it was not stable in the symptom network model, limiting our ability to draw conclusions about the relative importance of the variables included in that network. Additionally, closeness and betweenness were not reliably estimable in either of the networks. However, the latter two metrics are often unstable in psychometric networks, and their utility and interpretability have been questioned because they treat the associations as distances. Namely, they may be more appropriate for networks where the nodes correspond to, for example, individuals rather than psychological traits or other risk factors (Bringmann et al., 2019; Epskamp et al., 2018a).

5. Conclusion

Network analysis characterised how psychosocial risk factors interact with depression severity and core depressive symptoms in adolescence. Findings underscore the central roles of emotion regulation difficulties, perceived stress, and school environment in adolescents' depressive experiences, highlighting the increasing importance of school environment across this developmental period. Symptom-level analyses showed that low mood and anhedonia relate to distinct risk factors. Taken together, results add to the existing mental health literature in adolescent risk and resilience by specifically pointing to the value of interventions that target stress reduction, strengthen emotion regulation capacities, and consider the evolving environmental pressures adolescents face, particularly within school settings. This study also provides a foundation for further building on existing work through more granular and longitudinal studies examining how specific risk factors relate to a range of depressive symptoms in adolescence, which may inform the development of more targeted, developmentally-sensitive mental health interventions for young people.

CRediT authorship contribution statement

Jure Bužinel: Writing – review & editing, Writing – original draft, Visualization, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Jessica Norman:** Writing – review & editing, Project administration, Conceptualization. **Jessica Hughes-Nind:** Writing – review & editing, Project administration, Conceptualization. **Christina Carlisi:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

We have no competing interests to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2026.122080.

Data availability

All data and analysis scripts are available through GitHub: https://github.com/jbuzinel/adolescencemhnetwork

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